CESC 410L Course Info for Fall 2026

1 General Info

- Class meeting time:
  - CESC 410L-01: Tu 5:15 pm to 8:15 pm
  - CESC 410L-02: Fr 5:15 pm to 8:15 pm
- Room: LB 373
- Instructor: Dr. Jianhua Liu, Associate Professor of Electrical and Computer Engineering
  - The instructor is responsible for training the TA of CESC 410L.
  - The teaching of CESC 410L is the responsibility of TA.
  - Please see the TAs first if you have CESC 410L related issues; they can refer you to the instructor if needed.
- Instructor's contact info and office hours: Same as CESC 410
- TA's office hours: TBD

2 Prerequisites and Corequisites

- Prerequisite: CESC 315
- Corequisite: CESC 410

3 Catalog Course Description

Introduction to real-time digital signal processing (DSP) principles, complementing CEC 410; development and analysis of discrete-time signals using MATLAB; generation of discrete-time waveforms and execution of Discrete-Time Fourier Transform (DTFT) and Fast Fourier Transform (FFT) operations in MATLAB to analyze signal spectra; demonstrations of Fourier Transform properties; lowpass and bandpass sampling techniques; hands-on experience with filter design concepts using MATLAB Simulink.

Note: We use Python instead of MATLAB for Fall 2025.

4 Learning Outcomes

The goal of this class is to provide a solid understanding of the fundamentals of discrete-time signals and systems.

After taking this course, students will be able to do the following:

1. Simulate discrete-time signals using tools such as MATLAB.
2. Compute Fourier transforms of simulated signals.
3. Evaluate filter performance through simulation.
4. Design signal processing experiments using simulation tools.
5. Validate sampling techniques with simulated data.|

5 Teamwork

Students are encouraged to work in teams of two members (there can be ONE team with three members if we have an odd number of students in a section) for the weekly labs. If you **really** prefer to work alone, it is ok. Yet, those working in groups will have the priority of getting help from TAs. Each student working in teams needs to work with ***different*** team members—**we will rebuild teams every two labs, starting from Lab 3**. Each team will submit one report and code. However for those labs that we collect ABET artifacts, every student will have to submit their own report and code; the ABET artifacts will have to be independent work.

6 Required Textbook and Class notes

No book is needed. Lab handouts are posted on the CESC 510 Canvas site.

7 Required Hardware

- A laptop computer running Windows, macOS, or Linux.

8 Lab Contents and Meeting Schedule

We will have weekly labs unless otherwise specified. Students are required to meet once a week in the lab as scheduled for your section.

Many of the labs will involve programming, and sometimes it can take longer than planned time to finish. As such, the labs are organized in a flexible manner. Each lab will have ONE week to finish. Please submit the report and code of the old lab before coming to the lab meeting of the new one.

The lab time in the following schedule is the time to start the new lab not the time for submitting the lab.

Lab	Sec 2, Fri	Sec 1, Tue
Lab 0	8/28/26	9/1/26
Lab 1	9/4/26	9/8/26
Lab 2	9/11/26	9/15/26
Lab 3	9/18/26	9/22/26
Lab 4	9/25/26	9/29/26
No lab	10/2/26	10/6/26
Lab 5	10/9/26	10/13/26
No lab	10/16/26	10/20/26
Lab 6	10/23/26	10/27/26
Lab 7	10/30/26	11/3/26
No lab	11/6/26	11/10/26

Lab	Sec 2, Fri	Sec 1, Tue
Lab 8	11/13/26	11/17/26
Lab Final	11/20/26	12/1/26

For the final Lab Exam, it should be finished during the lab meeting time.

9 Assessment of Learning Outcomes

9.1 Weekly Labs

Although you are required to submit lab reports, the report will not be assessed before the ATs can see your live demo of the lab. The live demo happens at the beginning of each lab meeting.

9.2 Final Lab Exam

To make sure everyone can do their work independently, we have a final Lab Exam, as scheduled above. Narrowed-down problems will be posted ahead of time so that you can prepare.

10 Grading Weighting and Letter Grade Assignment

Weekly labs will add up to 60 points. The lab final exam will be weighted at 40 points. The total will be 100 points.

Grading Scale: 90+ => A, 80+ => B, 70+ => C, 60+ => D, 60- => F.

11 Other Class Policies

11.1 AI Use Policy for This Course

See the same policy in the Course Info for CEC 410.

11.2 Meeting Time Policies

Try your best to finish the lab during the scheduled meet time. Stay in the lab even if you cannot finish the lab assignments. The lab meet time is the best time to get help from lab TAs.

11.3 Missing Labs or Late Reports

If you cannot make it to the lab for the live demo, you need to submit a video demo before the start of the lab meeting. Everyone can submit up to TWO video demos. Missing submission of lab reports will receive automatically zero for that lab; no resubmission is allowed.

Note that late reports will not be accepted.

11.4 Academic Integrity

ERAU is committed to intellectual integrity and considers academic dishonesty a very serious offense. Such offenses include cheating (accepting or giving unauthorized assistance in preparing assignments), or plagiarism (taking the ideas, writings, and/or work of another and representing them as one's own without appropriate acknowledgment). A student who cheats or commits any form of academic fraud will be subject to a failing grade for the course. In addition, the incident will be reported to the Dean of Students. If any other academic integrity violations have been documented, the student will be recommended for dismissal.

12 Report Format

Your report shall include five sections: header, introduction, narrative, code snippet, and discussions and results, as shown below. For the narrative part, use Times Roman, size 12, single spacing; for the code part, use Currier New, size 12, single spacing.

A full submission includes:

- A soft copy of the report (electrical version) in PDF.
- Well-commented source code in a ZIP file. The ZIP file should contain the entire **cleaned** project which is directly buildable; do not include built results. For this, you need to clean the project in CubeIDE as instructed.

Below is a **Template for Lab Report**:

12.1 Header

Include the following information:

- CESC 410L Section number
- Lab number and Lab title
- Lab start date: MM/DD/YY
- Report data: MM/DD/YY
- Names of team members

12.2 Introduction

Briefly introduce the lab assignments in your own words. Include research done in preparation.

12.3 Narrative

Discuss how the procedure was different for you. Did you have to modify anything? Did you have any problems/bugs? How were they resolved? Did you do research to fix a problem? Talk about it here.

12.4 Code Snippet

Only include in the report the snippets of source code that you have added, modified, or commented. When you take screenshots, try not to have dark background—we may have to print out the reports and we don't want to waste too much ink.

Include the file name where the code are from.

12.5 Discussions and Results

This section should include your takeaway from the lab. What did you learn? How can you apply it in the future?

- Write down the answer for each question in the handouts if applicable.
- Summarize the lab and write down your results if applicable.